

# From Alligators to All In - Generalized Predator-Prey Problems and Guided Online Research

Wael El Khateeb (Wael.ElKhateeb@rockets.utoledo.edu),  
Steven J. Miller ([sjm1@williams.edu](mailto:sjm1@williams.edu))

with  
Jasmine Pham ([jpham9@huskers.unl.edu](mailto:jpham9@huskers.unl.edu)),  
Jessie Wang ([jw1648@scarletmail.rutgers.edu](mailto:jw1648@scarletmail.rutgers.edu)),

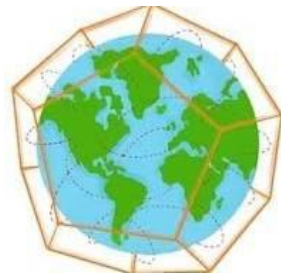
**AMS Contributed Paper Session on Mathematical Biology**



[https://web.williams.edu/Mathematics/sjmiller/public\\_html/math/talks/talks.html](https://web.williams.edu/Mathematics/sjmiller/public_html/math/talks/talks.html)

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# Polymath Junior Program



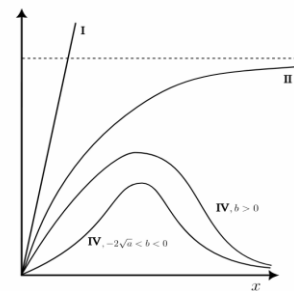
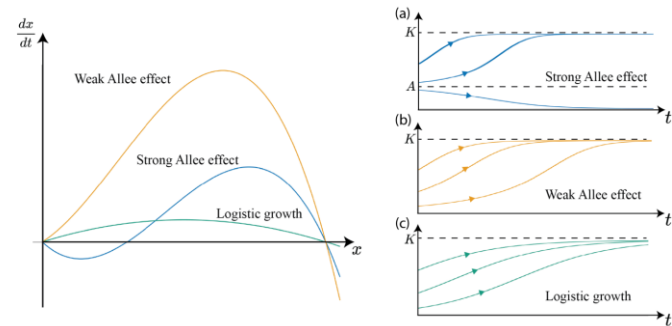
- Provide research opportunities to undergraduates.
- Online, runs in the spirit of the [Polymath Project](#).
- Projects run by researcher with experience in undergraduate mentoring.
- Most 15-25 students, a main mentor, grad students / postdocs assisting.



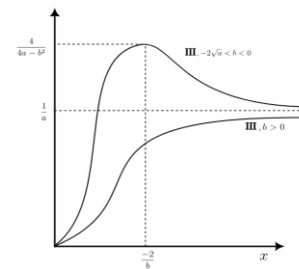
# Extensions of PP models

$$\begin{cases} \frac{dx}{dt} = \alpha x - mxy, \\ \frac{dy}{dt} = cxy - dy. \end{cases}$$

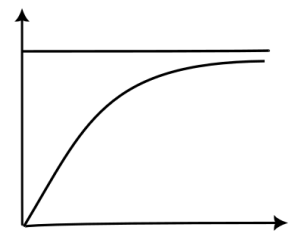
$$\begin{cases} \frac{dx}{dt} = G_1(x) - \alpha_1 F(x)y - H_1(x), \\ \frac{dy}{dt} = -G_2(y) + \alpha_2 F(x)y - H_2(y). \end{cases}$$



(a) Holling Type I, II, and IV



(b) Holling Type III



# Background (Biology)

1800s to mid-1900s: Intense hunting for leather and meat.

1950s to 1960s: Populations crash; poaching persists; weak enforcement.

1967: Listed as endangered in the United States.

1973: Endangered Species Act enabled.

1970s: States close hunting, protect nests, begin science-based management.

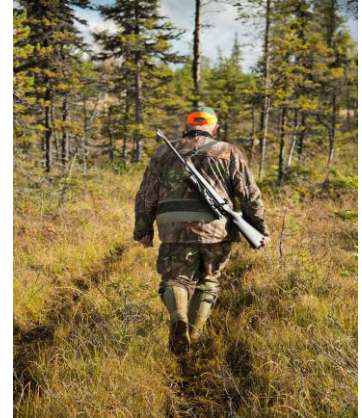


# Background (Biology)

Late 1970s to 1980s: Rebound via protection, anti-poaching, wetland conservation.

1987: Delisted federally; managed as “threatened due to similarity of appearance.”

FWC lottery permits; two alligators per permit.





# Burmese Pythons

- Intentional releases by owners who could no longer keep large snakes.
- Accidental escapes from pet stores, breeders, and private collections.
- Facility damage during Hurricane Andrew (1992) likely released additional snakes into the Everglades.

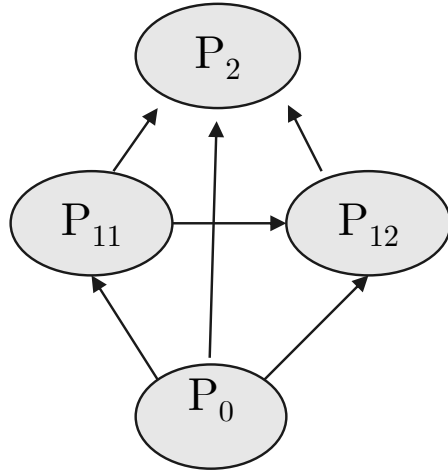


# Ecological implications



- Sharp declines of mid-sized mammals in parts of the Everglades (raccoons, opossums, marsh rabbits)
- Pressure on threatened species that nest or forage on the ground
- Competition with native apex predators for prey, altering food webs

## Model Illustrated:



$P_0$  - Racoon/prey population

$P_{11}$  - Juvenile Alligator population

$P_{12}$  - Adult Alligator population

$P_2$  - Burmese Python population



# Initial Model

$$f_m(P_n) = P_n$$

$$H_2(P_2) = h_2$$

$$H_3(P_3) = h_3 P_3$$

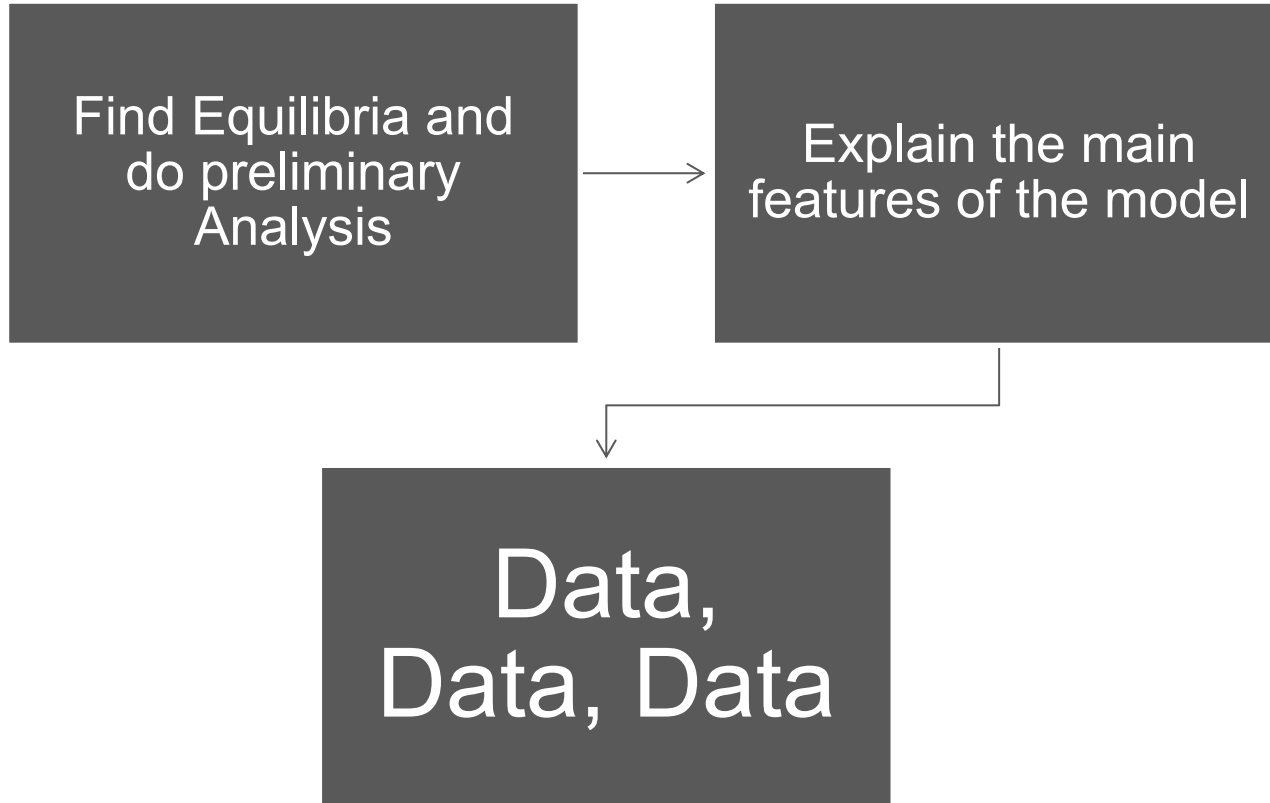
$$\begin{cases} \frac{dP_0}{dt} &= f_1(P_0) - m_{20}f_2(P_0)\bar{P}_{12} - m_{30}f_3(P_0)P_2 \\ \frac{dP_{11}}{dt} &= c_{20}f_2(P_0)P_{12} - m_{31}f_4(P_{11})P_3 - d_1P_{11} \\ \frac{dP_{12}}{dt} &= d_1P_{11} - d_2P_{12} - H_2(P_{12}) \\ \frac{dP_2}{dt} &= c_{30}f_3(P_0)P_2 + c_{31}f_4(P_{11})P_2 - d_3P_2 - H_3(P_2) \end{cases}$$

$h_2 \Rightarrow \text{Saturated}$

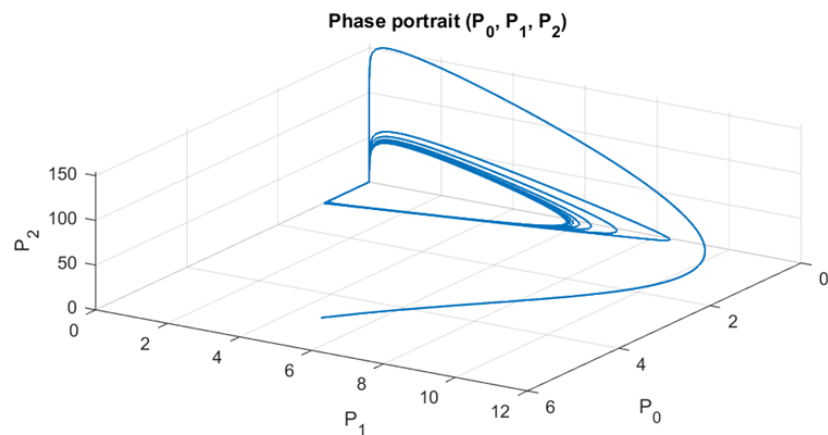
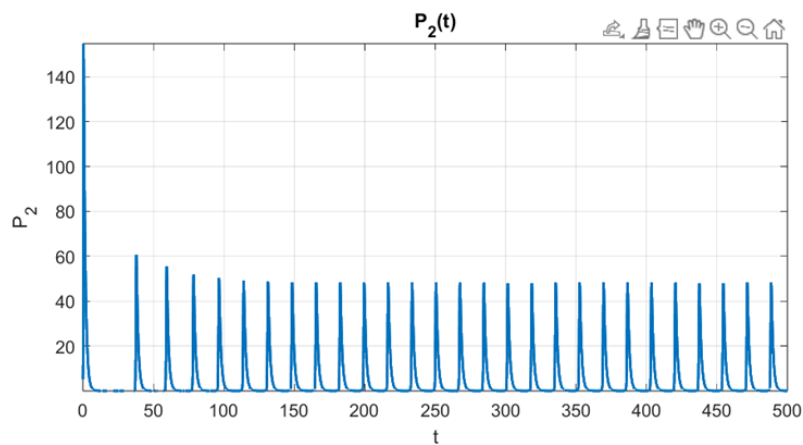
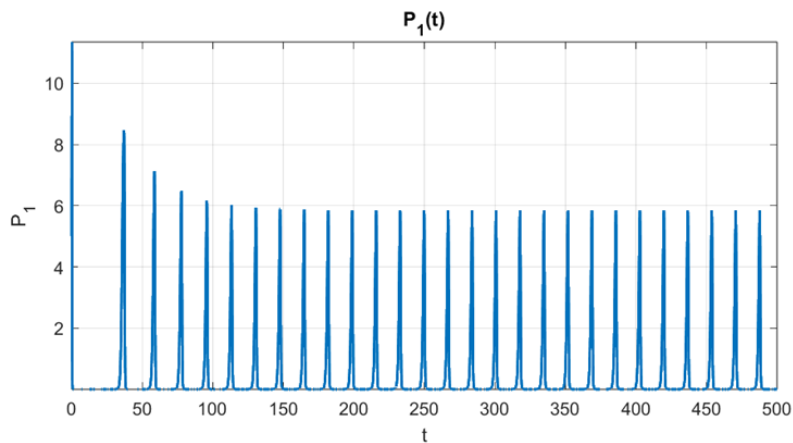
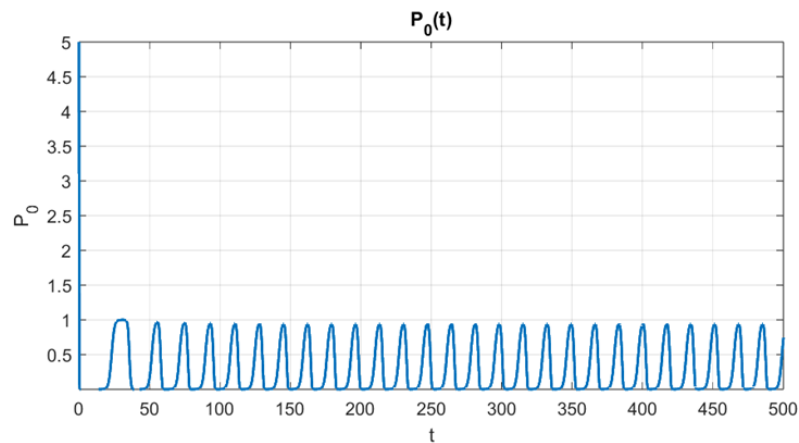
$$\begin{cases} \frac{dP_0}{dt} &= rP_0 - m_{20}P_0P_{12} - m_{30}P_0P_2 \\ \frac{dP_{11}}{dt} &= c_{20}P_0P_{12} - m_{31}P_{11}P_3 - d_{11}P_{11} \\ \frac{dP_{12}}{dt} &= d_{12}P_{11} - d_2P_{12} - h_2 \\ \frac{dP_2}{dt} &= c_{30}P_0P_2 + c_{31}P_2P_{11} - d_3P_2 - h_3P_2 \end{cases}$$

$$\begin{cases} \frac{dP_0}{dt} &= rP_0 \left(1 - \frac{P_0}{K}\right) - m_{10}P_0P_1 - m_{20}P_0P_2, \\ \frac{dP_1}{dt} &= c_{10}P_0P_1 - d_1P_1 - \frac{h_1P_1}{a + P_1} - m_{31}P_1P_2, \\ \frac{dP_2}{dt} &= c_{20}P_0P_2 + c_{21}P_1P_2 - d_2P_2 - h_2P_2. \end{cases}$$

# Ultimate goal: Policy making



# Presence of Limit Cycles



# Bogdanov Takens Bifurcation:

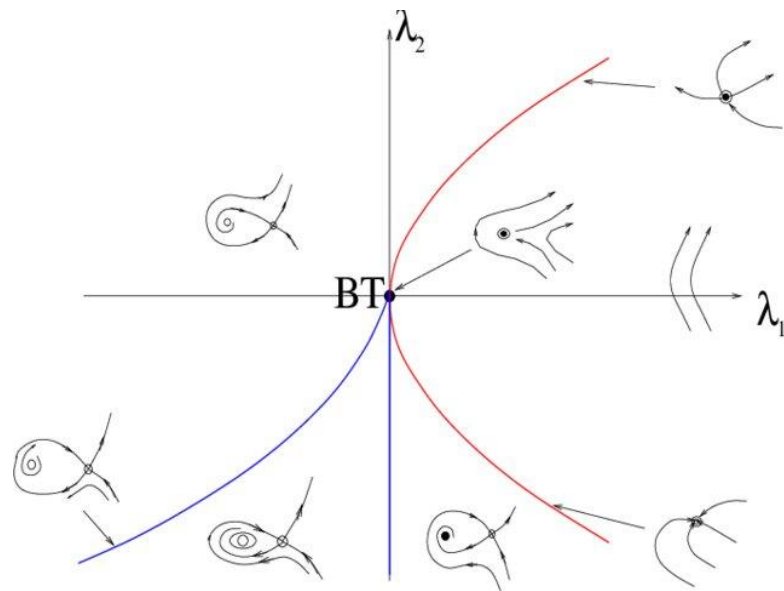
$$\dot{x} = f(x, \mu), \quad x \in \mathbb{R}^2, \quad \mu = (\mu_1, \mu_2)$$

A Bogdanov Takens Bifurcation occurs when the system has an equilibrium point  $x^*$  such that the Jacobian  $Df(x^*, \mu^*)$  has a double zero eigenvalue:

$$\text{tr}(Df) = 0, \quad \det(Df) = 0, \quad Df \neq 0,$$

And  $Df(x^*, \mu^*)$  has a single Jordan block (geometric multiplicity = 1). The system can then be transformed to the normal form:

$$\begin{cases} \dot{u} = v, \\ \dot{v} = \beta_1 + \beta_2 u + u^2 + s uv. \end{cases}$$



Steps followed:

1. **Shift the equilibrium:** Find the equilibrium  $P^* = (P_0^*, P_1^*, P_2^*)$  and shift variables:

$$u = P_0 - P_0^*, \quad v = P_1 - P_1^*, \quad w = P_2 - P_2^*$$

so that the system becomes  $\dot{z} = F(z)$  with  $F(0) = 0$ .

2. **Linearize and identify eigen-directions:** Compute the Jacobian  $J = DF(0)$ . At the BT point:  $\text{spec}(J) = \{0, 0, \lambda_s\}$  with  $\lambda_s < 0$ . The eigenvectors corresponding to zero eigenvalues define the center subspace.

3. **Center manifold reduction (3D  $\rightarrow$  2D):** There exists a smooth invariant manifold  $w = h(u, v)$  with  $h(0) = 0$ ,  $Dh(0) = 0$ . Substituting  $w = h(u, v)$  reduces the system to:

$$\dot{u} = \tilde{F}_1(u, v), \quad \dot{v} = \tilde{F}_2(u, v)$$

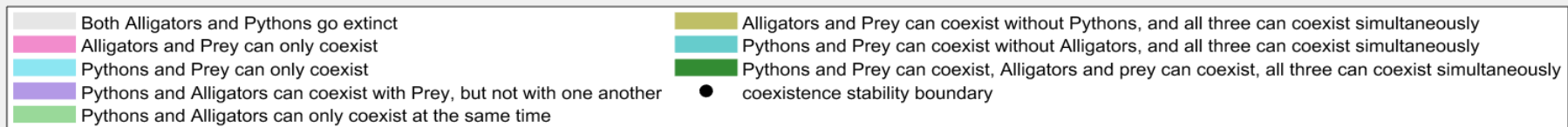
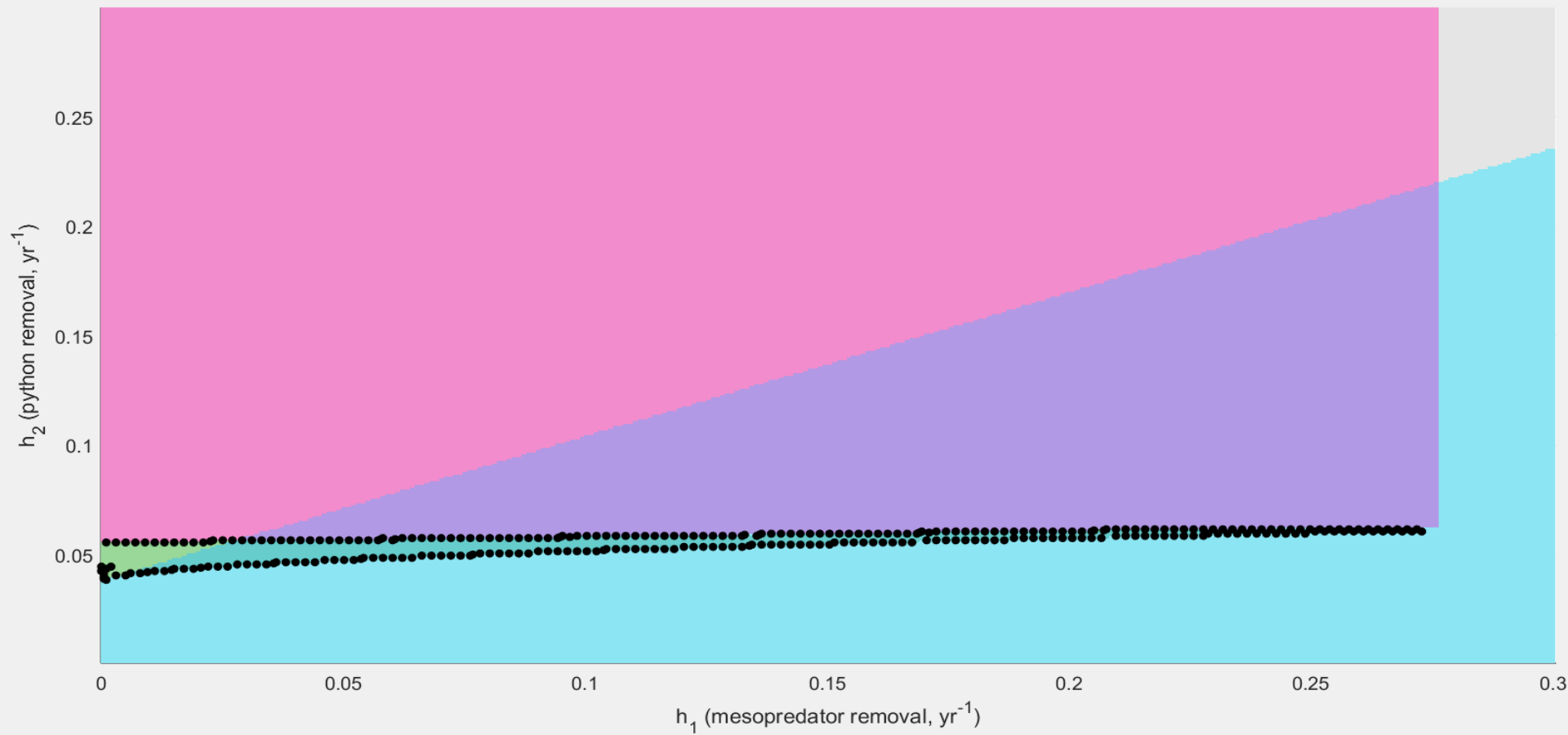
4. **Reduced dynamics:** Expand  $\tilde{F}_1, \tilde{F}_2$  in a Taylor series near the origin and retain quadratic terms:

$$\dot{u} = v + \mathcal{O}(2), \quad \dot{v} = au^2 + buv + \dots$$

5. **Transform to BT normal form:** Apply smooth coordinate and parameter changes to obtain:

$$\boxed{\dot{x} = y, \quad \dot{y} = \beta_1 + \beta_2 x + ax^2 + bxy + \mathcal{O}(3)}$$





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Thank you!

Steven J Miller  
Professor  
Williams College



Wael El Khateeb  
Ph.D. Candidate in Applied Math  
The University of Toledo



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